

Dynamic Modeling of Price-Volume Histories and Empirical Analysis of Prediction Markets as Bayesian Inverse Models

Forrest Gao, Nate Ly

Introduction

Prediction markets have become increasingly popular both as a source of forecasting and trading. It is estimated that prediction market volumes grew fourfold in 2025 alone [4]. This paper focuses on the forecasting component of prediction markets. Specifically, whether market prices and trading activity genuinely reflect an aggregation of public and private information about a binary event. Much of the existing literature approaches this question through structural equilibrium models [1], in which prices emerge from rational agent behavior. Recently, Ciani et al. (2026) proposed a Bayesian inverse framework that casts outcome inference as a marginal likelihood problem over a parametric model of trader types [3]. Given that prediction markets have matured into actively traded financial instruments, we test several of the modeling assumptions from Ciani et al. (2026) against real Polymarket data, and extend their framework to address two practically important concerns: endogenous volume dynamics and temporal dependence in trading activity, through a series of progressively richer Markov models.

Prior Literature

Ciani et al. (2026) formulate prediction markets as Bayesian inverse problems [3]. To infer a binary market outcome $Y \in \{0, 1\}$ from an observed history $h_T = (\Delta x_{1:T}, v_{1:T})$, they model the log-odds increments $(\Delta x_{1:T})$ conditionally upon the volume sequence $(v_{1:T})$, which is treated as a fixed design. To account for unobserved market participant behavior, they introduce a nuisance parameter $\Theta := (\omega, \gamma, \theta_{1:K})$ representing the dynamics of K latent trader types of informed traders (whose price increments drift toward the true outcome), uninformed noise traders (who are outcome-independent), and adversarial or manipulative traders (whose cause increments to drift away from the truth).

The generative model captures K latent trader types via the nuisance parameter $\Theta := (\omega, \gamma, \theta_{1:K})$. Volume-dependent softmax-gating yields the probability that a price increment was generated by trader type k , written as $\rho_k(v; \omega; \gamma)$. Type-specific increment laws $f_{k,y}(\delta x \mid v, \theta_k)$, are the conditional distribution of increments given a trader regime and conditioned on the volume and type-specific parameters. The stepwise mixture likelihood $f_y(\delta x \mid v, \Theta)$, derived from LOTP, is constructed using the ρ weighted sum of the increment laws. Through the key central assumption of conditional independence for price increments, the joint likelihood $p(\Delta x_{1:T} \mid v_{1:T}, Y = y, \Theta)$ factorizes into the product of the mixture likelihoods described above.

Finally, by placing a prior Π on the nuisance parameter Θ , the authors derive the joint posterior for (Y, Θ) . After integrating the nuisance structure, a marginal likelihood $m_y(h)$ for each outcome $y \in \{0, 1\}$ is obtained. It turns out inference reduces to evaluating a marginal Bayes factor $BF_T(h) := \frac{m_1(h)}{m_0(h)}$. Through this, the authors show that the log posterior odds update is purely additive and expressed as the sum of the log prior odds ratio and the log Bayes factor.

To explore the identifiability of this Bayesian problem, the authors define an outcome separation gap, which measures how close in distribution (via KL) a forward process under the wrong outcome, optimizing for nuisance structure, can get to the true forward process under y^*, θ^* . They were able to show that the log BF grows at least linearly in time with the asymptotic outcome separation gap. Because the log BF is roughly equal to the negative log posterior error, this means the posterior error decays at least exponentially.

Modeling Volume as Part of the Forward Process

1 Probabilistic Formulation

As noted in the prior literature section, the Ciani et al. (2026) paper treated volume as fixed-design. By Bayes Rule, we can write the joint posterior as:

$$P(Y = y, \Theta \in d\theta \mid h_{1:T}) \propto p(h_{1:T} \mid Y = y, \theta) P(Y = y, \Theta \in d\theta)$$

If we give the outcome of the market and the nuisance structure independent priors, the above is:

$$p(h_{1:T} \mid Y = y, \theta) \pi_0^y (1 - \pi_0)^{1-y} \Pi(d\theta) = p(\Delta x_{1:T}, v_{1:T} \mid Y = y, \theta) \pi_0^y (1 - \pi_0)^{1-y} \Pi(d\theta) \quad (1)$$

From here, the authors utilized the definition of conditional probability to factor out the volume term, which is then treated as a constant and discarded for the purposes of the inference problem.

1.1 Volume-Only Markov Process

Volume is shaped by information arrivals, including past price increments and volumes, and strategic participation in the market. By factoring out the volume path by the definition of conditional probability:

$$P(Y = y, \Theta \in d\theta \mid h_{1:T}) \propto \pi_0^y (1 - \pi_0)^{1-y} p(\Delta x_{1:T} \mid Y = y, \theta, v_{1:T}) p(v_{1:T} \mid Y = y, \theta) \Pi(d\theta) \quad (2)$$

Then, by the probability chain rule:

$$p(v_{1:T} \mid Y = y, \theta) = p(v_1 \mid Y = y, \theta) \prod_{t=2}^T p(v_t \mid v_{1:t-1}, Y = y, \theta)$$

Simulating traders reacting only to the most recent market activity, we assume volume is conditionally Markov; that is, the volume at time $t + 1$ depends only on the volume at time t and is conditionally independent of all prior history. Thus, by the Markov property:

$$p(v_{1:T} \mid Y = y, \theta) = p(v_1 \mid Y = y, \theta) \prod_{t=2}^T p(v_t \mid v_{t-1}, Y = y, \theta) = \prod_{t=1}^T p(v_t \mid v_{t-1}, Y = y, \theta)$$

By defining a baseline volume v_0 as a boundary condition, we can turn this sequence into a single product. Substituting this result into Equation 2 and using the same price increment law as Ciani et al. (2026), we get the joint posterior:

$$P(Y = y, \Theta \in d\theta \mid h_{1:T}) \propto \pi_0^y (1 - \pi_0)^{1-y} \prod_{t=1}^T f_y(\Delta x_t \mid v_t, \theta) \prod_{t=1}^T p(v_t \mid v_{t-1}, Y = y, \theta) \Pi(d\theta) \quad (3)$$

1.2 Jointly Markov Process

However, this volume-only Markov specification is undesirable because it introduces symmetry. A volume-only Markov model can render the inference non-identifiable unless one is willing to assume that the outcome directly affects volume dynamics, or one is willing to create an asymmetric prior for the different market outcomes. Thus, we propose that a “jointly-Markov” model is more natural, as then volume can respond to prior price movement as well as prior volume; when conditioned on both, we can have a more natural asymmetric law w.r.t $Y = 1$ and $Y = 0$. Under this, the first term in the LHS of (1) becomes:

$$p(h_{1:T} \mid Y = y, \theta) = p(h_1 \mid Y = y, \theta) \prod_{t=2}^T p(h_t \mid h_{1:t-1}, Y = y, \theta)$$

By modeling the joint price-volume history $h_t = (\Delta x_t, v_t)$ as a conditionally Markov process and introducing a sentinel boundary state v_0 , the joint likelihood factorizes as

$$p(h_{1:T} \mid Y = y, \theta) = \prod_{t=1}^T p(h_t \mid h_{t-1}, Y = y, \theta) = \prod_{t=1}^T p(\Delta x_t, v_t \mid \Delta x_{t-1}, v_{t-1}, Y = y, \theta)$$

Applying the definition of conditional probability, we arrive at:

$$\prod_{t=1}^T \left[p(v_t \mid \Delta x_{t-1}, v_{t-1}, Y = y, \theta) p(\Delta x_t \mid v_t, \Delta x_{t-1}, v_{t-1}, Y = y, \theta) \right] \quad (4)$$

Finally, the key assumption we make is as follows:

$$p(\Delta x_t | v_t, \Delta x_{t-1}, v_{t-1}, Y = y, \theta) = f_y(\Delta x_t | v_t, \Theta) \quad f_y(\Delta x_t | v, \Theta) := \sum_{k=1}^K \rho_k(v; \omega, \gamma) f_{k,y}(\Delta x_t | v_t, \theta_k)$$

That is, conditional on the current volume v_t , Δx_t is independent of the previous price increment and the previous volume (i.e. the previous price-volume state). *This is a simplifying modeling assumption:* as current volume sufficiently captures trading intensity, the lagged state only affects Δx_t via v_t modeling. Thus, we can utilize the mixture likelihood f_y similar to 3.6 in Cianci et al. (2026). Substituting this into (4), we arrive at:

$$p(h_{1:T} | Y = y, \theta) = \prod_{t=1}^T \left[p(v_t | \Delta x_{t-1}, v_{t-1}, Y = y, \theta) f_y(\Delta x_t | v_t, \theta) \right]$$

Finally, we can substitute this likelihood back into (1) to get the joint posterior:

$$P(Y = y, \Theta \in d\theta | h_{1:T}) \propto \left(\prod_{t=1}^T p(v_t | \Delta x_{t-1}, v_{t-1}, Y = y, \theta) f_y(\Delta x_t | v_t, \theta) \right) \pi_0^y (1 - \pi_0)^{1-y} \Pi(d\theta) \quad (5)$$

2 Parametric Instantiations of Volume-Modeling and Markov Structure

We use a framework of the factored joint likelihood:

$$p(\Delta x_{1:T}, v_{1:T} | Y = y, \theta) = \prod_{t=1}^T f_y(\Delta x_t | v_t, \theta_x) \cdot \prod_{t=1}^T p(v_t | v_{t-1}, \Delta x_{t-1}, Y = y, \theta_v)$$

This section serves to show the parametric choices for the volume transition $p(v_t | \cdot)$. Note: all random variables added to the nuisance structure are defined with priors, and the appended vector can be found in the appendix.

2.1 Transformed Volume and Autoregressive (AR) Backbone

We operate in the log space where the log-transformed volume $u_t = \log(1 + v_t)$, such that the density in the volume space is recovered through change of variables:

$$\log p(v_t | \cdot) = \log p(u_t | \cdot) - u_t$$

We define:

$$\mu_t^{\text{base}} = \alpha_v + \phi_v(u_{t-1} - \alpha_v) \quad \text{where} \quad u_t | h_{t-1}, \theta \sim \mathcal{N}(\mu_t^{\text{base}}, \sigma_v^2)$$

with long-run mean α_v , persistence $\phi_v = \tanh(z_{\phi_v})$, ensuring $\phi_v \in (-1, 1)$ and a stationary, outcome-neutral baseline.

2.2 Reversal

When a price drifts from the true outcome, informed traders identify a mispricing opportunity and increase their activity in the next period to push the price back towards the correct value, generating a reversal effect. Thus, we can capture this by adding a drift term that only activates after wrong-way price movements:

$$\mu_t^{\text{rev}} = \mu_t^{\text{base}} + \beta_{\text{rev}} \cdot [-(2y - 1)\Delta x_{t-1}]^+ \quad [x]^+ := \max(0, x) \quad u_t | h_{t-1}, Y = y, \theta \sim \mathcal{N}(\mu_t^{\text{rev}}, \sigma_v^2)$$

The expression $[-(2y - 1)\Delta x_{t-1}]^+$ isolates the magnitude of the market's deviation from the truth. Here, the $2y - 1$ term maps the binary outcome $\{0, 1\}$ to a directional sign $\{-1, 1\}$ to quantify the mispricing.

2.3 Reversal, Burst

Real-world markets exhibit volume spikes due to events like news and liquidity shocks, unrelated to the binary outcome. To account for this, we introduce a discrete mixture model using a heavy-tailed outcome-independent burst component:

$$p(u_t | h_{t-1}, Y = y, \theta) = (1 - \eta) \cdot \mathcal{N}(u_t; \mu_t^{\text{rev}}, \sigma_v^2) + \eta \cdot t_{\nu_b}(u_t; \mu_t^{\text{base}}, (\sigma_v + \sigma_b)^2)$$

With probability $1 - \eta$, the volume is drawn from the reversal channel, else drawn from a Student-t distribution centered around μ_t^{base} . This ensures generic volume spikes do not accumulate false evidence for either outcome.

2.4 Reversal, Momentum, Burst

To account for uninformed traders following trends, we add a momentum term, creating a directional mean replacing the above μ_t^{rev} with:

$$\mu_t^{\text{dir}} = \mu_t^{\text{base}} + \beta_{\text{rev}} \cdot [-(2y - 1)\Delta x_{t-1}]^+ + \beta_{\text{mom}} \cdot [(2y - 1)\Delta x_{t-1}]^+$$

However, this formulation introduces symmetry, where a history of wrong-way moves under $Y = 1$ driven by reversal is indistinguishable of right-way moves under $Y = 0$ driven by momentum, leading to non-identifiability

2.5 Gated Reversal, Momentum, Burst

To create identifiability, we use a volume-based filtering hypothesizing that reversals happen in low-volume environments, where informed traders advantage of the liquidity in the market to fix any pricing errors, while momentum is realized in high-volume environments of uninformed traders chasing visible order flow [2]. Similar to before, we replace the μ_t^{dir} term in the mixture with:

$$\mu_t^{\text{gated}} = \mu_t^{\text{base}} + \beta_{\text{rev}} \cdot [-(2y - 1)\Delta x_{t-1}]^+ \cdot g_{\text{low}}(u_{t-1}) + \beta_{\text{mom}} \cdot [(2y - 1)\Delta x_{t-1}]^+ \cdot g_{\text{high}}(u_{t-1})$$

where the gates are defined with a learned center c and slope $s > 0$:

$$g_{\text{high}}(u) = \sigma(s \cdot (u - c)) \quad g_{\text{low}}(u) = 1 - g_{\text{high}}(u)$$

This gating structure tries to break the direct reversal/momentum sign symmetry and improve identifiability by restricting the regime in which each mechanism can explain the next period's volume.

Theoretical Extensions Under New Volume and Price Modeling

1 Theorem 4.2 Under Jointly-Markov Assumption

As before, let $h_t = (\Delta x_t, v_t)$. For notational efficiency, let the one-step factor be:

$$g_y(h_t | h_{t-1}, \theta) := p(v_t | \Delta x_{t-1}, v_{t-1}, Y = y, \theta) f_y(\Delta x_t | v_t, \theta) \iff p(h_t | h_{t-1}, Y = y, \theta)$$

We now define Markov variants of the log-likelihood, marginal likelihood, and expected log-likelihood, similar to section 4.4.2 by Ciani et al. (2026) in their Gaussian-latent-type model. Let the log-likelihood be:

$$\ell_{y,T}^M(\theta) := \sum_{t=1}^T \log g_y(H_t | H_{t-1}, \theta) \iff \log(p(H_{1:T} | Y = y, \theta))$$

Here, the M superscript specifies a log-likelihood under the jointly-Markov assumption, distinguishing from the i.i.d. assumption in Ciani et al. (2026). Under the true law, we have $H_{1:T} \sim P_{y^*, \theta^*}^{(T), M}$. Then, we define the per-time expected log-likelihood under the jointly-Markov model to be:

$$\Lambda_{y,T}^M(\theta) := \frac{1}{T} \mathbb{E}_{P_{y^*, \theta^*}^{(T), M}} [\ell_{y,T}^M(\theta)]$$

The key claim to prove Theorem 4.2 in the reference paper was a Uniform Law of Large Numbers (ULLN):

$$\sup_{\theta \in \Theta} \left| \frac{1}{T} \ell_{y,T}^M(\theta) - \Lambda_{y,T}^M(\theta) \right| \rightarrow 0 \text{ a.s. under the true history-generating law}$$

This was treated as an assumption in the i.i.d. case. For the jointly-Markov case, this claim holds from a uniform ergodic theorem for Markov chains under stationarity of the true path process. Continuity of $\theta \mapsto \log(g_y(h_t | h_{t-1}, \theta))$, as well as assumptions 4.1 and 4.2 from the Ciani et al. (2026) paper, which are compactness and integrable envelope, respectively.

The outcome separation gap under the jointly-Markov assumption, analogous to 4.2 from Ciani et al. (2026) is:

$$\delta_T^M(y^*, \theta^*) := \inf_{\theta \in \Theta} \frac{1}{T} KL \left(P_{y^*, \theta^*}^{(T), M} \parallel P_{1-y^*, \theta}^{(T), M} \right)$$

Note that the KL-divergence is now over the joint path law of $(\Delta X_{1:T}, V_{1:T})$ rather than over only the log-odds price increments conditioned on fixed volume. Finally, notice that:

$$\Lambda_{y^*, T}^M(\theta^*) - \Lambda_{y, T}^M(\theta) = \frac{1}{T} KL \left(P_{y^*, \theta^*}^{(T), M} \parallel P_{y, \theta}^{(T), M} \right)$$

Notice that the above equation holds regardless of the specific likelihood factorization we choose (i.e. it works for both i.i.d. and jointly-Markov assumptions). Now, we mechanically have the pieces to 1. perform the same Laplace-type bounds for $\frac{1}{T} \log m_y^M(H_T)$ 2. relate suprema to δ_T^M , and 3. apply the same BF bound as the authors did for the i.i.d. case for our jointly-Markov assumption. Thus, we conclude that:

$$\delta_T^M - \epsilon \leq \frac{1}{T} \log BF_T^M(H_T) \leq \delta_T^M + \epsilon, \text{ for } y^* = 1$$

That is, under the jointly-Markov forward process, at sufficiently large T , the per-time log Bayes factor converges to the jointly-Markov outcome-separation gap, up to $\epsilon = o(1)$. Thus, we have extended the convergence guarantee of the log BF to jointly-Markov likelihood models, like the ones we described in the previous section.

2 SMC Algorithm Weight Update Under Jointly-Markov Assumption

In our empirical analysis, we use the Sequential Monte Carlo (SMC) algorithm for estimating the marginal likelihood under the jointly-Markov assumption, i.e. $m_1^M(h)$ and $m_0^M(h)$, where h is the cumulative history up to time T . The main SMC algorithm outline is described in *Algorithm 1* of the Cianci et al. paper. Note that, for the parametric instantiations described above, additional priors are needed; these are detailed in the appendix. Under our model, there is a change to step 4, for computing the incremental weights. In general, the factor used in the weight update step is the incremental likelihood ratio. This is:

$$u_{y,t}(\theta) = \frac{p(h_{1:t} | Y = y, \theta)}{p(h_{1:t-1} | Y = y, \theta)}$$

Then, the weight update for the i th particle is:

$$\tilde{w}_t^{(i)} = w_{t-1}^{(i)} \cdot u_{y,t}(\theta^{(i)})$$

We can simplify the weight update $u_{y,t}$ via a one-step chain rule on the numerator:

$$u_{y,t}(\theta) = \frac{p(h_{1:t-1} | Y = y, \theta) \cdot p(h_t | h_{1:t-1}, Y = y, \theta)}{p(h_{1:t-1} | Y = y, \theta)} = p(h_t | h_{1:t-1}, Y = y, \theta)$$

Applying Markov Property under the jointly-Markov assumption, the above is $p(h_t | h_{t-1}, Y = y, \theta)$, which is exactly $g_y(h_t | h_{t-1}, \theta)$. Thus, our new update step is:

$$\tilde{w}_t^{(i)} = w_{t-1}^{(i)} \cdot g_y(h_t | h_{t-1}, \theta^{(i)})$$

Finally, for step 11, we do an MCMC move in $\theta^{(i)}$ targeting $\pi_t = \Pi(\theta | \Delta x_{1:t}, v_{1:t}, y)$. By Bayes Rule and independence of nuisance structure prior:

$$\pi_t \propto p(h_{1:t} | Y = y, \theta) \Pi(\theta | Y = y) = p(h_{1:t} | Y = y, \theta) \Pi(\theta)$$

Empirical Analysis On Market Data

In the Cianci et al. (2026) paper, the authors focused on validating their theoretical results on identifiability, posterior concentration, and stability under perturbations for their Gaussian-latent-type model. To do so, they generated data under their “mock-market” forward process for the Gaussian-latent-type model and performed inference over that data using the same model class. We were interested, as an empirical exercise, to see how inference over real market data using the various models would perform.

1 Empirical Test Procedure

Our empirical procedure involved the following steps:

1. We analyzed 1000 recently resolved, binary Polymarket markets. Inclusion criteria required a clear outcome, a minimum volume of 10,000, and at least 200 trades. Price and volume data were aggregated into 30-minute intervals, retaining the final $T = 500$ periods (10.4 days) before resolution.
2. For each model, we constructed a likelihood and prior that together form an “inverse problem” object. The likelihood class supplies the one-step factor for updating the SMC weights. We ran the SMC marginal likelihood estimator separately under $Y = 0$ and $Y = 1$ using 1500 particles, applying two RW-Metropolis rejuvenation steps whenever we hit the resampling condition, which is on line 8 of the algorithm in Cianci et al. (2026).

- For each combination of resolved market and model, we cached the full posterior trace $(\pi_t)_{t \in [T]}$, the cumulative log BF trace, per-step log marginal likelihood increments, and effective sample size (ESS). We then computed empirical metrics. Let $y \in \{0, 1\}$ be the ground-truth resolved outcome and π_1 be the model posterior probability of $Y = 1$. Log loss is computed as $-\log(\pi_1^y(1-\pi_1)^{1-y})$ and Brier score as $(\pi_1 - y)^2$. For the accuracy, we threshold at 0.5: if $\pi_1 \geq 0.5$, we treat this as our model classifying that the market resolves to “Yes” and vice versa for $\pi_1 < 0.5$. We define a model as being “overconfident and wrong” on a particular market if it assigned either greater than 90% to the posterior probability or less than 10% on the posterior probability, and it was incorrect relative to the true outcome.

2 Empirical Test Results

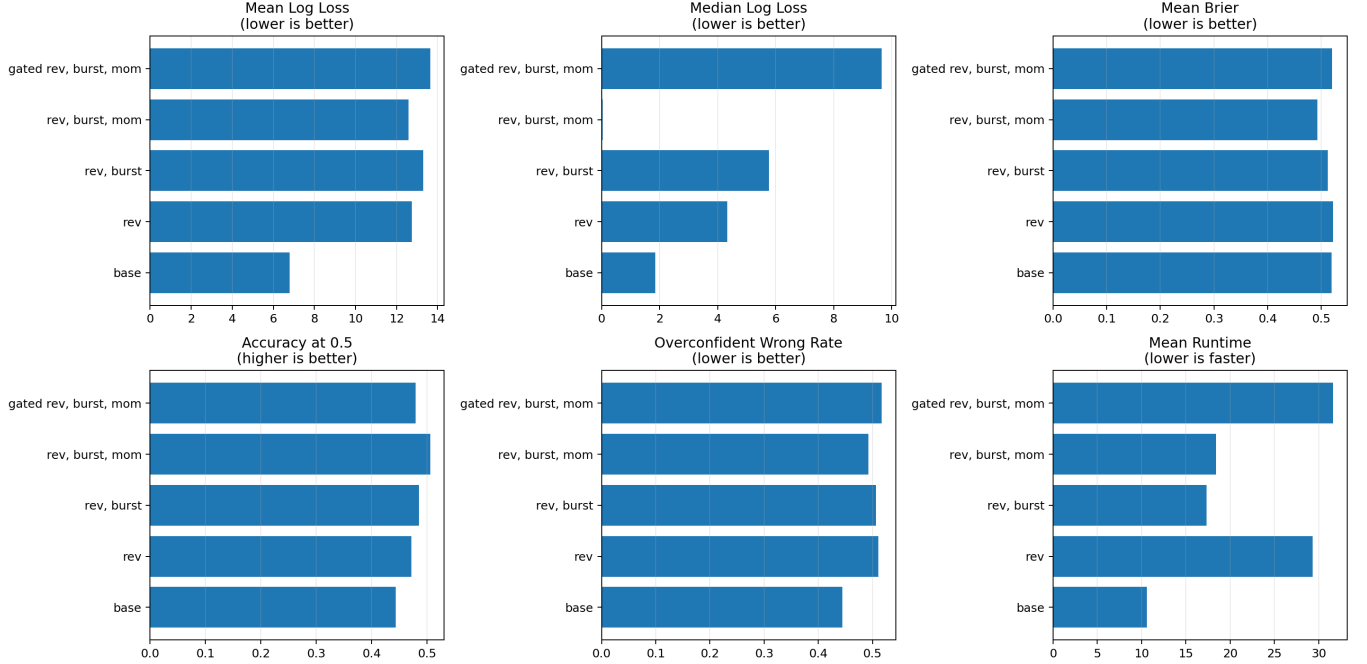


Figure 1: Results of various models under the jointly-Markov assumption when applied to inference over real Poly-market price-volume trajectories.

From the various results, every model performed poorly in absolute terms. For example, if we had proposed 0.5 probability at the end of each market the Brier Score would have been $(0.5)^2 = 0.25$, which is lower than the average Brier score of any of the models.

However, we do believe that this empirical exercise was not fruitless. In terms of absolute accuracy, the reversal, burst, momentum model performed marginally better than all of the other models, which may be surprising due to the theoretical symmetry between reversal and momentum under different market outcomes. Our main hypothesis for this was that flexible models perform better under real market data. As we will never be able to replicate the “true-market” generative model, all of the likelihood models are simply approximations. For example, the model with the volume-gate performs worse due to a strong inductive bias as it makes strict assumptions about how markets work. Although these may make sense from a market microstructure standpoint, they are certainly not representative of general market conditions, thus leading to worse performance than the ungated model due to this inflexibility.

Synthetic Validation of Forward Process

In this section, we empirically validate Theorem 4.2 extended to the jointly-Markov assumption. To examine the identifiability conditions, we simulated the forward data-generating process under known parameters, which can be found in the appendix for reproducibility. By generating synthetic price-volume histories where the true market resolution is fixed (e.g., $Y = 1$), we can directly measure the posterior concentration rate—specifically, how rapidly the log posterior error decays as the time horizon T increases.

1 Synthetic Experiment Procedure

For these experiments, we reused our prior, model and inference classes from the market data test, which were responsible for computing the incremental likelihoods and performing the SMC algorithm respectively. Now, for each of our prior/model pairs, we created a sampler class to draw a trajectory and performed the following experiment:

1. Draw 1000 sample histories of length $T = 500$ from the sampler class. Denote the i th history as $H^{(i)}$.
2. For each $H^{(i)}$, perform inference over the whole path via SMC, caching the the prefix results for each $T \in \{10, 25, 50, 100, 200, 500\}$. Note that simply taking the prefix yields length- T draws that are from the appropriate length- T forward process, but one could also sample 1000 different trajectories for each T . Compute the log-posterior error at each time T .
3. Compute the estimated outcome-separation gap under jointly-Markov assumption, denoted as $\hat{\delta}_T^M$. To do this, we draw another 1000 sample histories of length $T = 500$ from the forward sampler class, and construct an empirical distribution $\hat{P}_{y^*, \theta^*}^{(T), M}$. Then, we compute the empirical outcome separation gap via:

$$\hat{\Lambda}_{y^*, T}^M(\theta^*) - \sup_{\theta \in \Theta} \hat{\Lambda}_{1-y^*, T}^M(\theta) \quad (6)$$

Here, $\hat{\Lambda}$ represents the empirical per-time expected log-likelihood, with the expectation taken with respect to $\hat{P}_{y^*, \theta^*}^{(T), M}$. Further details on estimating the supremum are discussed in the appendix.

2 Synthetic Experiment Results

Summary plots of our synthetic results are displayed in *Figure 2*. The plots show the concentration of $\log(1 - \pi_T)$, i.e. the log posterior error (recall that the outcomes were fixed to 1).

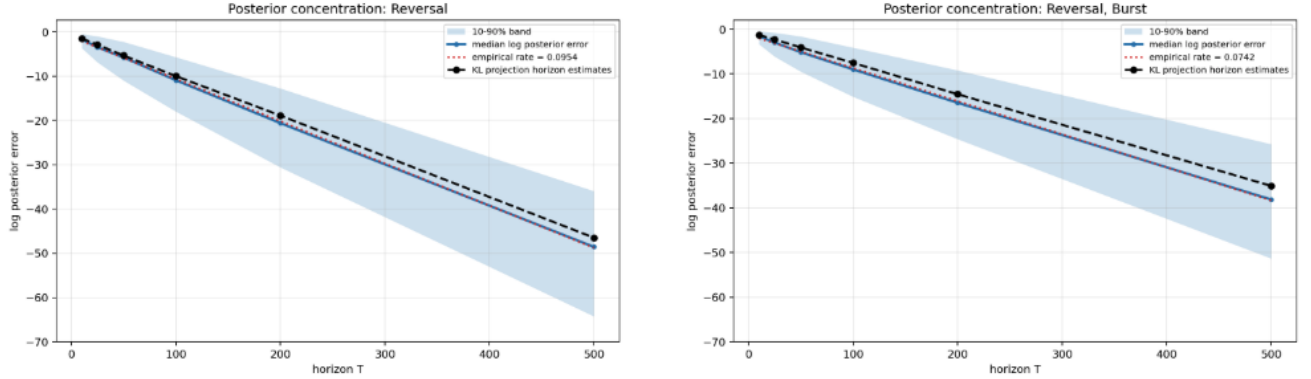


Figure 2: Identifiability tests on Jointly-Markov Models: Reversal (Left) vs. Reversal, Burst (Right)

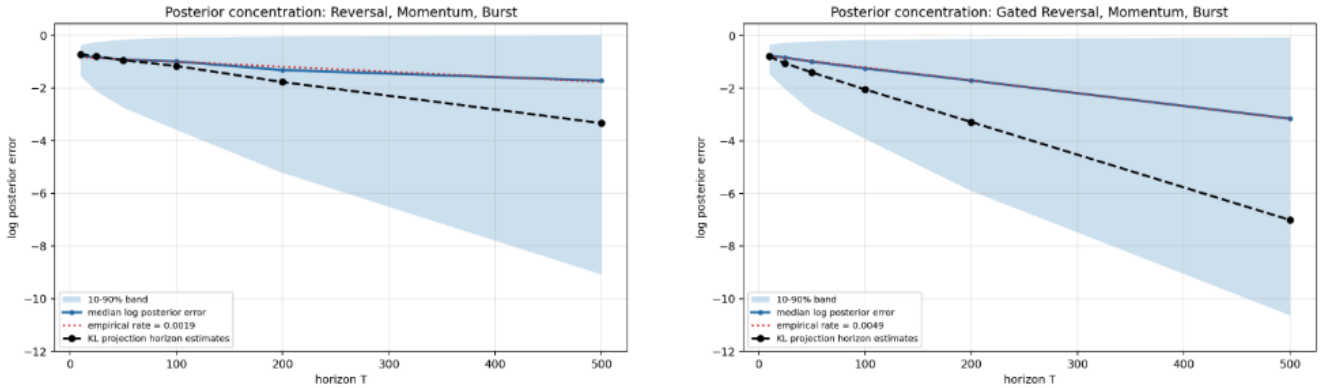


Figure 3: Reversal, Momentum, Burst (Left) vs. Gated Reversal, Momentum, Burst (Right)

Across the synthetic experiments, models with more parameters exhibit wider posterior error bands and slower median concentration rates. This can be explained through the structure, where a larger nuisance parameter space gives the

wrong-outcome model more degrees of freedom to mimic histories generated under the true outcome, which compresses the outcome separation gap. However, since real markets operate under unknown dynamics, strong theoretical convergence in synthetic settings does not guarantee optimal performance on real-world data. It is also important to note that models that better capture true market dynamics are harder to identify and require much larger sample sizes to produce stable Bayes factors.

Figure 2 shows the results from the Reversal model and the Reversal, Burst model. Both track the proposed theoretical rate (decreasing linearly in time with a slope equal to the outcome-separation gap) quite closely, as evidenced by the closeness of the dashed and solid black lines. In addition, their 10–90% bands are quite narrow. Notice that the Reversal, Burst model converges at a slower rate compared to the Reversal model, consistent with our hypothesis above that convergence is worse under more parameters, as there are more degrees of freedom to create potential non-identifiability.

Figure 3 shows the result comparisons between the Reversal, Momentum, Burst model, and the Gated Reversal, Momentum, Burst model. As discussed previously, the Reversal, Momentum, Burst model has a symmetry in it: reversal under a market outcome of 1 generates volume in the same way that momentum under a market outcome of 0 does, if the previous price movement is the same. We can see that upon introducing the volume-gating for reversal and momentum, both the theoretical rate and the empirical rate of concentration improve.

Interestingly, despite superior theoretical identifiability, recall that the gated volume model underperformed the non-gated version on real market data. This highlights a modeling tradeoff: one can impose strong market assumptions to create a more structured model, leading to better theoretical results, at the cost of potential inductive bias. If one’s theory on the market is incorrect, then the performance on real market data can be expected to be worse.

Conclusion

This paper extends the Bayesian inverse problem framework of Cianci et al. (2026) by treating volume as part of the forward process rather than as a fixed design sequence. We describe a jointly-Markov likelihood for price-volume histories, in which the current volume may depend on the previous price-volume state and the outcome of the market, while the current price increment is, under a simplifying assumption, the same as the increment law from the original paper’s framework. Furthermore, we show that the same SMC marginal-likelihood estimator applies after replacing the original incremental weight by the jointly Markov one-step likelihood factor.

We then study several parametric instantiations. The reversal model captures informed traders correcting mispricings after wrong-way price moves. Adding a burst parameter allows for outcome-independent microstructure effects and news events to generate volume spikes without needing to contribute to evidence on Y . The momentum component captures scenarios in which uninformed traders add positions to follow trends after informed traders have made positions. Finally, gating the reversal and momentum effects to specific volume regimes reduces symmetry and improves outcome separation.

Several limitations motivate future work. First, richer model classes exhibit worse posterior concentration and wider uncertainty bands, even as symmetry is resolved. The SMC particle count required for adequate posterior convergence scales with the parameter dimension; thus, a higher-parameter volume model is more susceptible to weight degeneracy than a reversal-only baseline. Further extensions should tune the particle counts, rejuvenation frequency, and proposal steps as a function of model dimension rather than simply using the same parameters across all model tests.

Next, based on the results, we see an issue of model overconfidence, seen in section 4 of the appendix. In the chart, we see the jointly-Markov model instantiations all saturate too early, meaning further large price swings closer to the market resolution have little effect on updating the posterior, leading to poor performance. This could be resolved through regularization methods during the updates, such as a dampening factor, RMS normalization, imposing bounds on update magnitudes with clipping, or adding a KL penalty.

Finally, extending latent mixture modeling to the volume process could further improve identifiability. Time-varying mixture weights could distinguish low-activity regimes (dominated by patient, informed traders) from high-activity regimes (dominated by momentum traders), allowing both channels to share information and unify market inferences.

References

- [1] David Easley, Soeren Hvidkjaer, and Maureen O'Hara. Is information risk a determinant of asset returns? *The Journal of Finance*, 57(5):2185–2221, 2002.
- [2] David Easley, Nicholas M. Kiefer, Maureen O'Hara, and Joseph B. Paperman. Liquidity, information, and infrequently traded stocks. *The Journal of Finance*, 51(4):1405–1436, 1996.
- [3] Juan P. Madrigal-Cianci, Camilo Monsalve Maya, and Lachlan Breakey. Prediction markets as bayesian inverse problems: Uncertainty quantification, identifiability, and information gain from price–volume histories under latent types, 2026. Accessed: 2026-05-04.
- [4] MEXC. FalconX: Prediction market volumes grew 4x to \$64B in 2025, 2026. Accessed: 2026-05-04.

Appendix

1 Priors for Parametric Model Instantiations

The base Gaussian latent-type prior is inherited from the original price-increment model described by Ciani et al. (2026). Recall that for all of the jointly-Markov models, we use a transformed volume:

$$u_t = \log(1 + v_t)$$

The shared volume priors are:

$$\begin{aligned}\alpha_v &\sim \mathcal{N}(0, 5^2), \\ z_{\phi_v} &\sim \mathcal{N}(0, 1), \quad \phi_v = \tanh(z_{\phi_v}), \\ \sigma_v &\sim \text{HalfNormal}(1)\end{aligned}$$

The reversal model adds a nonnegative reversal coefficient:

$$\beta_{rev} \sim \text{HalfNormal}(1)$$

The reversal, burst model adds an outcome-independent burst channel with:

$$\begin{aligned}\eta &\sim \text{Beta}(1, 9), \\ \sigma_b &\sim \text{HalfNormal}(3), \\ \nu_b - 2 &\sim \Gamma(2, 2)\end{aligned}$$

The reversal, momentum, burst model replaces the single reversal coefficient with two coefficients:

$$\begin{aligned}\beta_{rev} &\sim \text{HalfNormal}(1), \\ \beta_{mom} &\sim \text{HalfNormal}(1)\end{aligned}$$

Finally, the priors for the gated version of the reversal, momentum, burst model is:

$$\begin{aligned}c &\sim \mathcal{N}(0, 5^2), \\ s &\sim \text{HalfNormal}(2)\end{aligned}$$

2 True Parameters for Synthetic Experiments

All synthetic histories are generated under $Y = 1$ using the following fixed parameter vectors θ^* . The base price-increment parameters are the same as in Ciani et al. (2026).

The shared volume parameters, common to all jointly-Markov specifications, are:

Parameter	Description	Value
α_v	long-run mean log-volume	$\log 2$
ϕ_v	AR persistence	0.70
σ_v	innovation scale	0.25

Table 1: Shared volume parameters θ^*

The model-specific parameters are:

Model	Parameter	Description	Value
Reversal	β_{rev}	reversal coefficient	0.45
Reversal, Burst	β_{rev}	reversal coefficient	0.45
	η	burst mixture weight	0.08
	σ_b	burst extra scale	0.70
	ν_b	burst degrees of freedom	5.0
Rev, Momentum, Burst	β_{rev}	reversal coefficient	0.45
	β_{mom}	momentum coefficient	0.20
	η	burst mixture weight	0.08
	σ_b	burst extra scale	0.70
	ν_b	burst degrees of freedom	5.0
Gated Reversal, Momentum, Burst	β_{rev}	reversal coefficient	0.45
	β_{mom}	momentum coefficient	0.20
	c	gate center	$\log 2$
	s	gate slope	3.0
	η	burst mixture weight	0.08
	σ_b	burst extra scale	0.70
	ν_b	burst degrees of freedom	5.0

Table 2: Model-specific volume parameters θ^*

3 Outcome Separation Gap Estimation

Recall that we estimate the outcome separation gap via (6). For the first term, we estimate the per-time expected log likelihood via the empirical distribution of the forward process. First, we sample $S = 1000$ histories. Let the i th history be denoted by $H_{1:T}^{(i)}$. Then, the empirical expected log likelihood is:

$$\hat{\Lambda}_{y^*,T}^M(\theta^*) = \frac{1}{S} \sum_{i=1}^S \frac{1}{T} \log \left(p \left(H_{1:T}^{(i)} \mid Y = y^*, \theta^* \right) \right)$$

The second term we needed to compute was the optimal opposite-outcome per-time log likelihood:

$$\sup_{\theta} \hat{\Lambda}_{1-y^*,T}^M(\theta) = \sup_{\theta} \frac{1}{S} \sum_{i=1}^S \frac{1}{T} \log \left(p \left(H_{1:T}^{(i)} \mid Y = 1 - y^*, \theta \right) \right)$$

We approximately solved this problem via `scipy.optimize`. Using *random candidates* = 2048, *local restarts* = 8, *optimizer* = *Powell*, *maxiter* = 100, and $z \in [-8, 8]^d$. The random search samples candidate θ 's from the model prior, maps them to unconstrained coordinates z , clips to $[-8, 8]$, scores all candidates, and runs Powell optimization from the best 8 candidates for each horizon. The same S histories are reused across all candidate θ 's for each model.

4 Model Overconfidence under Real Market Data

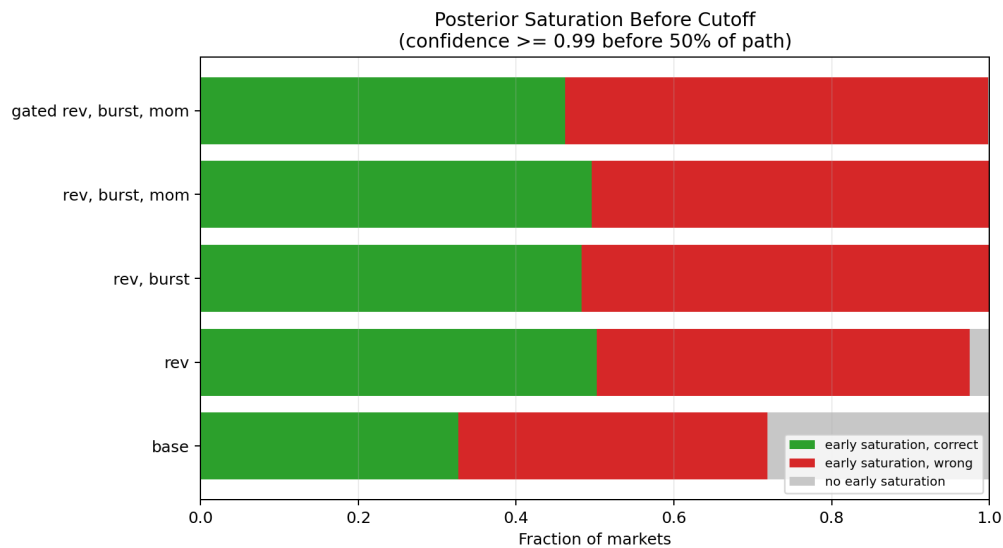


Figure 4: For each of the jointly-Markov instantiations, we determined the proportion of real-world markets under which the model saturated early (i.e. $> 99\%$ or $< 1\%$) before the halfway point of the market.

5 Github Repository

<https://github.com/natejly/pred-mkts-dynamic-price-volume>